Output tables for the test of Multiple comparisons.

June 16, 2025

1 Average rankings of Friedman test

Average ranks obtained by applying the Friedman procedure

Friedman statistic considering reduction performance (distributed according to chi-square with 17 degrees of freedom: 261.055556.

P-value computed by Friedman Test: 1.2711720565050655E-10.

Algorithm	Ranking
BCA	3.6944
BN	6.3056
CLARA	11
CP	8.2222
CA	6.6389
FF	10.6111
KM	6.6944
NC	11.9583
ND	9.9167
PAM	11.5278
P	11.5694
RE	17.5556
RSC	12.3472
SC	13.0139
S	7.75
SW	8.9167
TCC	9.9444
UPGMC	3.3333

Table 1: Average Rankings of the algorithms

2 Post hoc comparisons

Results achieved on post hoc comparisons for $\alpha = 0.05$, $\alpha = 0.10$ and adjusted p-values.

2.1 P-values for $\alpha = 0.05$

Nemenyi's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.000327.
Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.000538.
Shaffer's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.000327.

i	algorithms	$z = (R_0 - R_i)/SE$	p	Holm	Shaffer
153	RE vs. UPGMC	11.302676	0	0.000327	0.000327
152	BCA vs. RE	11.015694	0	0.000329	0.000368
151	BN vs. RE	8.940593	0	0.000331	0.000368
150	CA vs. RE	8.675687	0	0.000333	0.000368
149	KM vs. RE	8.631536	0	0.000336	0.000368
148	RE vs. S	7.792665	0	0.000338	0.000368
147	SC vs. UPGMC	7.693325	0	0.00034	0.000368
146	CP vs. RE	7.417381	0	0.000342	0.000368
145	BCA vs. SC	7.406343	0	0.000345	0.000368
144	RSC vs. UPGMC	7.163512	0	0.000347	0.000368
143	BCA vs. RSC	6.87653	0	0.00035	0.000368
142	RE vs. SW	6.865493	0	0.000352	0.000368
141	NC vs. UPGMC	6.854455	0	0.000355	0.000368
140	BCA vs. NC	6.567473	0	0.000357	0.000368
139	P vs. UPGMC	6.545397	0	0.00036	0.000368
138	PAM vs. UPGMC	6.512284	0	0.000362	0.000368
137	BCA vs. P	6.258415	0	0.000365	0.000368
136	BCA vs. PAM	6.225302	0	0.000368	0.000368
135	CLARA vs. UPGMC	6.092849	0	0.00037	0.000413
134	ND vs. RE	6.070773	0	0.000373	0.000413
133	RE vs. TCC	6.048698	0	0.000376	0.000413
132	BCA vs. CLARA	5.805867	0	0.000379	0.000413
131	FF vs. UPGMC	5.783791	0	0.000382	0.000413
130	FF vs. RE	5.518885	0	0.000385	0.000413
129	BCA vs. FF	5.496809	0	0.000388	0.000413
128	BN vs. SC	5.331243	0	0.000391	0.000413
127	TCC vs. UPGMC	5.253978	0	0.000394	0.000413
126	ND vs. UPGMC	5.231903	0	0.000397	0.000413
125	CLARA vs. RE	5.209827	0	0.0004	0.000413
124	CA vs. SC	5.066336	0	0.000403	0.000413
123	KM vs. SC	5.022185	0.000001	0.000407	0.000413
122	BCA vs. TCC	4.966996	0.000001	0.00041	0.000413
121	BCA vs. ND	4.944921	0.000001	0.000413	0.000413
120	BN vs. RSC	4.80143	0.000002	0.000417	0.000417
119	PAM vs. RE	4.790392	0.000002	0.00042	0.000463
118	P vs. RE	4.757279	0.000002	0.000424	0.000463
117	CA vs. RSC	4.536523	0.000006	0.000427	0.000463
116	BN vs. NC	4.492372	0.000007	0.000431	0.000463
115	KM vs. RSC	4.492372	0.000007	0.000435	0.000463
114	NC vs. RE	4.448221	0.000009	0.000439	0.000463
113	SW vs. UPGMC	4.437183	0.000009	0.000442	0.000463
112	CA vs. NC	4.227466	0.000024	0.000446	0.000463
111	KM vs. NC	4.183315	0.000029	0.00045	0.000463
110	SC vs. S	4.183315	0.000029	0.000455	0.000463
109	BN vs. P	4.183315	0.000029	0.000459	0.000463
108	BCA vs. SW	4.150201	0.000033	0.000463	0.000463
107	BN vs. PAM	4.150201	0.000033	0.000467	0.000472
106	RE vs. RSC	4.139164	0.000035	0.000472	0.000472
105	CA vs. P	3.918408	0.000089	0.000476	0.000476
104	CA vs. PAM	3.885295	0.000102	0.000481	0.000515
103	CP vs. UPGMC	3.885295	0.000102	0.000485	0.000515
102	KM vs. P	3.874257	0.000107	0.00049	0.000515
101	KM vs. PAM	3.841144	0.000122	0.000495	0.000515
100	CP vs. SC	3.808031	0.00014	0.0005	0.000515
99	BN vs. CLARA	3.730766	0.000191	0.000505	0.000515
98	RSC vs. S	3.653502	0.000259	0.00051	0.000515
97	RE vs. SC	3.609351	0.000307	0.000515	0.000515
96	BCA vs. CP	3.598313	0.00032	0.000521	0.000532
95	S vs. UPGMC	3.510011	0.000448	0.000526	0.000532
94	CLARA vs. CA	3.46586	0.000529	0.000532	0.000532
93	CLARA vs. KM	3.421709	0.000622	0.000538	0.000538
92	BN vs. FF	3.421709	0.000622	0.000543	0.000543
91	NC vs. S	3.344444	0.000824	0.000549	0.000549
90	CP vs. RSC	3.278218	0.001045	0.000556	0.000556
89	SC vs. SW	3.256142	0.001129	0.000562	0.000562
88	BCA vs. S	3.223029	0.001268	0.000568	0.000568
87	CA vs. FF	3.156802	0.001595	0.000575	0.000575
86	FF vs. KM	3.112651	0.001854	0.000581	0.000581
85	P vs. S	3.035387	0.002402	0.000588	0.000588
84	PAM vs. S	3.002273	0.00268	0.000595	0.000595
83	CP vs. NC	2.96916	0.002986	0.000602	0.000602
82	BN vs. TCC	2.891896	0.003829	0.00061	0.00061
81	BN vs. ND	2.86982	0.004107	0.000617	0.000617
80	RSC vs. SW	2.726329	0.006404	0.000625	0.000625
79	KM vs. UPGMC	2.67114	0.007559	0.000633	0.000633
78	CP vs. P	2.660102	0.007812	0.000641	0.000641
77	CA vs. TCC	2.626989	0.008614	0.000649	0.000649
76	CP vs. PAM	2.626989	0.008614	0.000658	0.000658
75	CA vs. UPGMC	2.626989	0.008614	0.000667	0.000667
74	CA vs. ND	2.604914	0.00919	0.000676	0.000676
73	CLARA vs. S	2.582838	0.009799	0.000685	0.000685
72	KM vs. TCC	2.582838	0.009799	0.000694	0.000694
71	KM vs. ND	2.560763	0.010444	0.000704	0.000704
70	ND vs. SC	2.461423	0.013839	0.000714	0.000714
69	SC vs. TCC	2.439347	0.014714	0.000725	0.000725
68	NC vs. SW	2.417272	0.015637	0.000735	0.000735
67	BCA vs. KM	2.384158	0.017118	0.000746	0.000746
66	BN vs. UPGMC	2.362083	0.018173	0.000758	0.000758
65	BCA vs. CA	2.340007	0.019283	0.000769	0.000769

2.2 P-values for $\alpha = 0.10$

Nemenyi's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.000654.

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.001124.

Shaffer's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.000654.

i	algorithms	$z = (R_0 - R_i)/SE$	p	Holm	Shaffer
153	RE vs. UPGMC	11.302676	0	0.000654	0.000654
152	BCA vs. RE	11.015694	0	0.000658	0.000735
151	BN vs. RE	8.940593	0	0.000662	0.000735
150	CA vs. RE	8.675687	0	0.000667	0.000735
149	KM vs. RE	8.631536	0	0.000671	0.000735
148	RE vs. S	7.792665	0	0.000676	0.000735
147	SC vs. UPGMC	7.693325	0	0.00068	0.000735
146	CP vs. RE	7.417381	0	0.000685	0.000735
145	BCA vs. SC	7.406343	0	0.00069	0.000735
144	RSC vs. UPGMC	7.163512	0	0.000694	0.000735
143	BCA vs. RSC	6.87653	0	0.000699	0.000735
142	RE vs. SW	6.865493	0	0.000704	0.000735
141	NC vs. UPGMC	6.854455	0	0.000709	0.000735
140	BCA vs. NC	6.567473	0	0.000714	0.000735
139	P vs. UPGMC	6.545397	0	0.000719	0.000735
138	PAM vs. UPGMC	6.512284	0	0.000725	0.000735
137	BCA vs. P	6.258415	0	0.00073	0.000735
136	BCA vs. PAM	6.225302	0	0.000735	0.000735
135	CLARA vs. UPGMC	6.092849	0	0.000741	0.000826
134	ND vs. RE	6.070773	0	0.000746	0.000826
133	RE vs. TCC	6.048698	0	0.000752	0.000826
132	BCA vs. CLARA	5.805867	0	0.000758	0.000826
131	FF vs. UPGMC	5.783791	0	0.000763	0.000826
130	FF vs. RE	5.518885	0	0.000769	0.000826
129	BCA vs. FF	5.496809	0	0.000775	0.000826
128	BN vs. SC	5.331243	0	0.000781	0.000826
127	TCC vs. UPGMC	5.253978	0	0.000787	0.000826
126	ND vs. UPGMC	5.231903	0	0.000794	0.000826
125	CLARA vs. RE	5.209827	0	0.0008	0.000826
124	CA vs. SC	5.066336	0	0.000806	0.000826
123	KM vs. SC	5.022185	0.000001	0.000813	0.000826
122	BCA vs. TCC	4.966996	0.000001	0.00082	0.000826
121	BCA vs. ND	4.944921	0.000001	0.000826	0.000826
120	BN vs. RSC	4.80143	0.000002	0.000833	0.000833
119	PAM vs. RE	4.790392	0.000002	0.00084	0.000926
118	P vs. RE	4.757279	0.000002	0.000847	0.000926
117	CA vs. RSC	4.536523	0.000006	0.000855	0.000926
116	BN vs. NC	4.492372	0.000007	0.000862	0.000926
115	KM vs. RSC	4.492372	0.000007	0.00087	0.000926
114	NC vs. RE	4.448221	0.000009	0.000877	0.000926
113	SW vs. UPGMC	4.437183	0.000009	0.000885	0.000926
112	CA vs. NC	4.227466	0.000024	0.000893	0.000926
111	KM vs. NC	4.183315	0.000029	0.000901	0.000926
110	SC vs. S	4.183315	0.000029	0.000909	0.000926
109	BN vs. P	4.183315	0.000029	0.000917	0.000926
108	BCA vs. SW	4.150201	0.000033	0.000926	0.000926
107	BN vs. PAM	4.150201	0.000033	0.000935	0.000943
106	RE vs. RSC	4.139164	0.000035	0.000943	0.000943
105	CA vs. P	3.918408	0.000089	0.000952	0.000952
104	CA vs. PAM	3.885295	0.000102	0.000962	0.001031
103	CP vs. UPGMC	3.885295	0.000102	0.000971	0.001031
102	KM vs. P	3.874257	0.000107	0.00098	0.001031
101	KM vs. PAM	3.841144	0.000122	0.00099	0.001031
100	CP vs. SC	3.808031	0.00014	0.001	0.001031
99	BN vs. CLARA	3.730766	0.000191	0.00101	0.001031
98	RSC vs. S	3.653502	0.000259	0.00102	0.001031
97	RE vs. SC	3.609351	0.000307	0.001031	0.001031
96	BCA vs. CP	3.598313	0.00032	0.001042	0.001064
95	S vs. UPGMC	3.510011	0.000448	0.001053	0.001064
94	CLARA vs. CA	3.46586	0.000529	0.001064	0.001064
93	CLARA vs. KM	3.421709	0.000622	0.001075	0.001075
92	BN vs. FF	3.421709	0.000622	0.001087	0.001087
91	NC vs. S	3.344444	0.000824	0.001099	0.001099
90	CP vs. RSC	3.278218	0.001045	0.001111	0.001136
89	SC vs. SW	3.256142	0.001129	0.001124	0.001136
88	BCA vs. S	3.223029	0.001268	0.001136	0.001136
87	CA vs. FF	3.156802	0.001595	0.001149	0.001149
86	FF vs. KM	3.112651	0.001854	0.001163	0.001163
85	P vs. S	3.035387	0.002402	0.001176	0.001176
84	PAM vs. S	3.002273	0.00268	0.00119	0.00119
83	CP vs. NC	2.96916	0.002986	0.001205	0.001205
82	BN vs. TCC	2.891896	0.003829	0.00122	0.00122
81	BN vs. ND	2.86982	0.004107	0.001235	0.001235
80	RSC vs. SW	2.726329	0.006404	0.00125	0.00125
79	KM vs. UPGMC	2.67114	0.007559	0.001266	0.001266
78	CP vs. P	2.660102	0.007812	0.001282	0.001282
77	CA vs. TCC	2.626989	0.008614	0.001299	0.001299
76	CP vs. PAM	2.626989	0.008614	0.001316	0.001316
75	CA vs. UPGMC	2.626989	0.008614	0.001333	0.001333
74	CA vs. ND	2.604914	0.00919	0.001351	0.001351
73	CLARA vs. S	2.582838	0.009799	0.00137	0.00137
72	KM vs. TCC	2.582838	0.009799	0.001389	0.001389
71	KM vs. ND	2.560763	0.010444	0.001408	0.001408
70	ND vs. SC	2.461423	0.013839	0.001429	0.001429
69	SC vs. TCC	2.439347	0.014714	0.001449	0.001449
68	NC vs. SW	2.417272	0.015637	0.001471	0.001471
67	BCA vs. KM	2.384158	0.017118	0.001493	0.001493
66	BN vs. UPGMC	2.362083	0.018173	0.001515	0.001515
65	BCA vs. CA	2.340007	0.019283	0.001538	0.001538

2.3 Adjusted p-values

i	hypothesis	unadjusted p	p_{Neme}	p_{Holm}	p_{Shaf}	p_{Berg}
1	RE vs .UPGMC	0	0	0	0	0
2	BCA vs .RE	0	0	0	0	0
3	BN vs .RE	0	0	0	0	0
4	CA vs .RE	0	0	0	0	0
5	KM vs .RE	0	0	0	0	0
6	RE vs .S	0	0	0	0	0
7	SC vs .UPGMC	0	0	0	0	0
8	CP vs .RE	0	0	0	0	0
9	BCA vs .SC	0	0	0	0	0
10	RSC vs .UPGMC	0	0	0	0	0
11	BCA vs .RSC	0	0	0	0	0
12	RE vs .SW	0	0	0	0	0
13	NC vs .UPGMC	0	0	0	0	0
14	BCA vs .NC	0	0	0	0	0
15	P vs .UPGMC	0	0	0	0	0
16	PAM vs .UPGMC	0	0	0	0	0
17	BCA vs .P	0	0	0	0	0
18	BCA vs .PAM	0	0	0	0	0
19	CLARA vs .UPGMC	0	0	0	0	0
20	ND vs .RE	0	0	0	0	0
21	RE vs .TCC	0	0	0	0	0
22	BCA vs .CLARA	0	0.000001	0.000001	0.000001	0
23	FF vs .UPGMC	0	0.000001	0.000001	0.000001	0
24	FF vs .RE	0	0.000005	0.000004	0.000004	0
25	BCA vs .FF	0	0.000006	0.000005	0.000005	0
26	BN vs .SC	0	0.000015	0.000012	0.000012	0
27	TCC vs .UPGMC	0	0.000023	0.000019	0.000018	0
28	ND vs .UPGMC	0	0.000026	0.000021	0.00002	0
29	CLARA vs .RE	0	0.000029	0.000024	0.000023	0
30	CA vs .SC	0	0.000062	0.00005	0.000049	0
31	KM vs .SC	0.000001	0.000078	0.000063	0.000062	0
32	BCA vs .TCC	0.000001	0.000104	0.000083	0.000082	0
33	BCA vs .ND	0.000001	0.000117	0.000092	0.000092	0
34	BN vs .RSC	0.000002	0.000241	0.000189	0.000189	0
35	PAM vs .RE	0.000002	0.000255	0.000198	0.000189	0
36	P vs .RE	0.000002	0.0003	0.000232	0.000212	0
37	CA vs .RSC	0.000006	0.000875	0.000669	0.000618	0
38	BN vs .NC	0.000007	0.001078	0.000817	0.000761	0
39	KM vs .RSC	0.000007	0.001078	0.000817	0.000761	0
40	NC vs .RE	0.000009	0.001325	0.000987	0.000935	0
41	SW vs .UPGMC	0.000009	0.001394	0.00103	0.000984	0
42	CA vs .NC	0.000024	0.003616	0.002647	0.002552	0
43	KM vs .NC	0.000029	0.004396	0.003189	0.003103	0
44	SC vs .S	0.000029	0.004396	0.003189	0.003103	0
45	BN vs .P	0.000029	0.004396	0.003189	0.003103	0
46	BCA vs .SW	0.000033	0.005082	0.003588	0.003588	0
47	BN vs .PAM	0.000033	0.005082	0.003588	0.003588	0
48	RE vs .RSC	0.000035	0.005333	0.003695	0.003695	0
49	CA vs .P	0.000089	0.013638	0.009359	0.009359	0
50	CA vs .PAM	0.000102	0.015637	0.010629	0.009914	0
51	CP vs .UPGMC	0.000102	0.015637	0.010629	0.009914	0
52	KM vs .P	0.000107	0.016363	0.010909	0.010374	0
53	KM vs .PAM	0.000122	0.018737	0.012369	0.011879	0
54	CP vs .SC	0.00014	0.021432	0.014008	0.013588	0
55	BN vs .CLARA	0.000191	0.029207	0.018899	0.018517	0
56	RSC vs .S	0.000259	0.039579	0.025351	0.025093	0
57	RE vs .SC	0.000307	0.046966	0.029776	0.029776	0
58	BCA vs .CP	0.00032	0.049004	0.030748	0.030107	0
59	S vs .UPGMC	0.000448	0.068558	0.042568	0.04212	0
60	CLARA vs .CA	0.000529	0.080866	0.049683	0.049683	0
61	CLARA vs .KM	0.000622	0.09521	0.057873	0.057873	0
62	BN vs .FF	0.000622	0.09521	0.057873	0.057873	0
63	NC vs .S	0.000824	0.126145	0.075027	0.075027	0
64	CP vs .RSC	0.001045	0.159831	0.094018	0.091929	0
65	SC vs .SW	0.001129	0.172794	0.100514	0.099385	0
66	BCA vs .S	0.001268	0.19407	0.111622	0.111622	0
67	CA vs .FF	0.001595	0.24405	0.138773	0.133988	0
68	FF vs .KM	0.001854	0.283685	0.159457	0.155749	0
69	P vs .S	0.002402	0.367548	0.204193	0.201791	0
70	PAM vs .S	0.00268	0.409996	0.225096	0.225096	0
71	CP vs .NC	0.002986	0.456881	0.24785	0.244864	0
72	BN vs .TCC	0.003829	0.585875	0.313999	0.313999	0
73	BN vs .ND	0.004107	0.628379	0.332671	0.332671	0
74	RSC vs .SW	0.006404	0.97986	0.512345	0.512345	0
75	KM vs .UPGMC	0.007559	1.156589	0.597193	0.597193	0
76	CP vs .P	0.007812	1.195188	0.609312	0.609312	0
77	CA vs .TCC	0.008614	1.318004	0.663309	0.654695	0
78	CP vs .PAM	0.008614	1.318004	0.663309	0.654695	0
79	CA vs .UPGMC	0.008614	1.318004	0.663309	0.654695	0
80	CA vs .ND	0.00919	1.406031	0.680041	0.670851	0
81	CLARA vs .S	0.009799	1.499267	0.715336	0.715336	0
82	KM vs .TCC	0.009799	1.499267	0.715336	0.715336	0
83	KM vs .ND	0.010444	1.597973	0.741543	0.731099	0
84	ND vs .SC	0.013839	2.117325	0.968711	0.968711	0
85	SC vs .TCC	0.014714	2.251216	1.015254	1.015254	0
86	NC vs .SW	0.015637	2.392514	1.06334	1.06334	0
87	BCA vs .KM	0.017118	2.619091	1.146922	1.146922	0
88	BN vs .UPGMC	0.018173	2.780406	1.199391	1.199391	0
89	BCA vs .CA	0.019283	2.950356	1.253419	1.253419	0